A POINT LIES IN THE QUASI-DM LOCUS EXACTLY WHEN ITS AUTOMORPHISM GROUP IS LOCALLY QUASI-FINITE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The existence of a maximal open subscheme over which the automorphism group space is locally quasi-finite, with base change formation. Invoked at the start of the counted proof (source lines 4832-4836) as a cross-chapter pointer. Produces the open subscheme U' of the presentation whose invariant descent gives the quasi-DM locus.
- The correspondence between open substacks of a presentation and invariant open subschemes. Invoked inside the counted proof (source lines 4841-4844) as a cross-chapter pointer. Descends the open subscheme U' to an open substack of the stack.
- The characterization of the quasi-DM locus through the diagonal being locally quasi-finite. Invoked inside the counted proof (source lines 4849-4851). Turns local quasi-finiteness of the inertia over the descended substack into quasi-DM-ness.

Source reference. Original lemma lemma-diagonal-diagonal: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 3959–3959. Navigation pointer only: the counted text refers to the Stacks item lemma-diagonal-diagonal, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference spaces-groupoids-lemma-open-over-which-unramified-or-lqf: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 4840–4840. Navigation pointer only: the counted text refers to the Stacks item spaces-groupoids-lemma-open-over-which-unramified-or-lqf, which belongs to

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Source reference. Original reference stacks-properties-lemma-substacks-presentation: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 4851–4851. Navigation pointer only: the counted text refers to the Stacks item stacks-properties-lemma-substacks-presentation, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let $\mathcal{X}$ be an algebraic stack. Let $x \in |\mathcal{X}|$ correspond to $x : \mathrm{Spec}(k) \to \mathcal{X}$. Let G_x/k be the automorphism group algebraic space of x. Then*

(1) *x is in the DM locus of $\mathcal{X}$ if and only if $G_x \to \mathrm{Spec}(k)$ is unramified, and*
(2) *x is in the quasi-DM locus of $\mathcal{X}$ if and only if $G_x \to \mathrm{Spec}(k)$ is locally quasi-finite.*

Proof. Proof of (2). Choose a scheme U and a surjective smooth morphism $U \to \mathcal{X}$. Consider the fibre product

$$\begin{array}{ccc} G & \longrightarrow & \mathcal{I}_{\mathcal{X}} \\ \downarrow & & \downarrow \\ U & \longrightarrow & \mathcal{X} \end{array}$$

Recall that G is the automorphism group algebraic space of $U \to \mathcal{X}$. By Groupoids in Spaces, Lemma spaces-groupoids-lemma-open-over-which-unramified-or-lqf there is a maximal open subscheme $U' \subset U$ such that $G_{U'} \to U'$ is locally quasi-finite. Moreover, formation of U' commutes with arbitrary base change. In particular the two inverse images of U' in $R = U \times_{\mathcal{X}} U$ are the same open subspace of R (since after all the two maps $R \to \mathcal{X}$ are isomorphic and hence have isomorphic automorphism group spaces). Hence U' is the inverse image of an open substack $\mathcal{X}' \subset \mathcal{X}$ by Properties of Stacks, Lemma stacks-properties-lemma-substacks-presentation and we have a cartesian diagram

$$\begin{array}{ccc} G_{U'} & \longrightarrow & \mathcal{I}_{\mathcal{X}'} \\ \downarrow & & \downarrow \\ U' & \longrightarrow & \mathcal{X}' \end{array}$$

Thus the morphism $\mathcal{I}_{\mathcal{X}'} \to \mathcal{X}'$ is locally quasi-finite and we conclude that $\mathcal{X}'$ is quasi-DM by Lemma lemma-diagonal-diagonal part (5). On the other hand, if $\mathcal{W} \subset \mathcal{X}$ is an open substack which is quasi-DM, then the inverse image $W \subset U$ of $\mathcal{W}$ must be contained in U' by our construction of U' since $\mathcal{I}_{\mathcal{W}} = \mathcal{W} \times_{\mathcal{X}} \mathcal{I}_{\mathcal{X}}$ is locally quasi-finite over $\mathcal{W}$. Thus $\mathcal{X}'$ is the quasi-DM locus. Finally, choose a field extension K/k and a 2-commutative diagram

$$\begin{array}{ccc} \mathrm{Spec}(K) & \longrightarrow & \mathrm{Spec}(k) \\ \downarrow & & \downarrow{\scriptstyle x} \\ U & \longrightarrow & \mathcal{X} \end{array}$$

Then we find an isomorphism $G_x \times_{\mathrm{Spec}(k)} \mathrm{Spec}(K) \cong G \times_U \mathrm{Spec}(K)$ of group algebraic spaces over K. Hence G_x is locally quasi-finite over k if and only if $\mathrm{Spec}(K) \to U$ maps into U' (use the commutation of formation of U' and Groupoids

in Spaces, Lemma spaces-groupoids-lemma-open-over-which-unramified-or-lqf applied to $\mathrm{Spec}(K) \to \mathrm{Spec}(k)$ and G_x to see this). This finishes the proof of (2). The proof of (1) is exactly the same. □